

Norway

SLEIPNER

Volve F

F-10

F-10

Design: F-10

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-10					
Well Position	+N/-S	-3.91 m	Northing:	6,478,559.62 m	Latitude:	58° 26' 29.6816 N
	+E/-W	2.23 m	Easting:	435,052.25 m	Longitude:	1° 53' 15.0698 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-10				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2008	05.04.2009	-2.02	71.61	50,402

Design	F-10				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.91	2.23	124.70	

Survey Program	Date	21.02.2013			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.00	4,764.09	8 1/2" Keeper Gyro (F-10)	Keeper, cont	Gyro Tool from SDC	
4,777.63	5,300.07	8 1/2" MWD svy tied in to Last Gyro (F-10)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	-3.91	2.23	0.00	0.000	0.000	0.000	
150.00	0.02	28.15	150.00	-3.91	2.23	0.00	0.146	0.146	0.000	
160.00	0.05	27.73	160.00	-3.91	2.23	0.00	0.090	0.090	-1.260	
170.00	0.04	30.13	170.00	-3.90	2.24	0.00	0.031	-0.030	7.200	
180.00	0.03	35.69	180.00	-3.89	2.24	0.00	0.032	-0.030	16.680	
190.00	0.04	31.77	190.00	-3.89	2.24	0.00	0.031	0.030	-11.760	
200.00	0.04	37.25	200.00	-3.88	2.25	0.00	0.011	0.000	16.440	
210.00	0.06	67.69	210.00	-3.88	2.25	0.00	0.098	0.060	91.320	
220.00	0.14	85.78	220.00	-3.87	2.27	0.01	0.255	0.240	54.270	
230.00	0.20	111.58	230.00	-3.88	2.30	0.04	0.287	0.180	77.400	

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Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
240.00	0.27	125.12	240.00	-3.90	2.33	0.08	0.267	0.210	40.620
250.00	0.28	127.84	250.00	-3.93	2.37	0.13	0.049	0.030	8.160
260.00	0.28	131.59	260.00	-3.96	2.41	0.18	0.055	0.000	11.250
270.00	0.30	129.68	270.00	-3.99	2.45	0.23	0.067	0.060	-5.730
280.00	0.31	129.81	280.00	-4.03	2.49	0.28	0.030	0.030	0.390
290.00	0.34	129.07	290.00	-4.06	2.53	0.34	0.091	0.090	-2.220
300.00	0.41	128.66	300.00	-4.10	2.58	0.40	0.210	0.210	-1.230
310.00	0.55	123.69	310.00	-4.15	2.65	0.49	0.438	0.420	-14.910
320.00	0.84	119.22	320.00	-4.22	2.76	0.61	0.884	0.870	-13.410
330.00	1.20	117.73	330.00	-4.30	2.91	0.78	1.083	1.080	-4.470
340.00	1.54	116.39	339.99	-4.41	3.13	1.02	1.024	1.020	-4.020
350.00	2.10	114.55	349.99	-4.54	3.41	1.33	1.689	1.680	-5.520
360.00	2.64	117.59	359.98	-4.73	3.78	1.74	1.663	1.620	9.120
370.00	2.89	119.53	369.97	-4.96	4.21	2.22	0.801	0.750	5.820
380.00	3.46	117.59	379.95	-5.22	4.69	2.77	1.740	1.710	-5.820
390.00	4.06	115.82	389.93	-5.52	5.28	3.42	1.833	1.800	-5.310
400.00	4.32	115.51	399.90	-5.83	5.94	4.14	0.783	0.780	-0.930
410.00	4.53	115.75	409.87	-6.17	6.63	4.91	0.632	0.630	0.720
420.00	4.78	116.07	419.84	-6.52	7.36	5.71	0.754	0.750	0.960
430.00	4.98	115.78	429.81	-6.89	8.13	6.55	0.605	0.600	-0.870
440.00	5.08	115.36	439.77	-7.27	8.92	7.41	0.320	0.300	-1.260
450.00	5.39	115.50	449.73	-7.66	9.74	8.31	0.931	0.930	0.420
460.00	5.79	115.11	459.68	-8.08	10.62	9.28	1.205	1.200	-1.170
470.00	6.04	114.60	469.62	-8.51	11.56	10.29	0.766	0.750	-1.530
480.00	6.31	114.83	479.57	-8.96	12.54	11.35	0.813	0.810	0.690
490.00	6.62	114.05	489.50	-9.43	13.56	12.46	0.967	0.930	-2.340
500.00	6.99	113.89	499.43	-9.91	14.64	13.62	1.111	1.110	-0.480
510.00	7.14	114.00	509.36	-10.41	15.77	14.83	0.452	0.450	0.330
520.00	7.38	113.71	519.28	-10.92	16.92	16.07	0.728	0.720	-0.870
530.00	7.79	114.26	529.19	-11.46	18.13	17.37	1.249	1.230	1.650
540.00	8.40	114.62	539.09	-12.04	19.41	18.75	1.836	1.830	1.080
550.00	8.86	114.55	548.98	-12.66	20.78	20.23	1.380	1.380	-0.210
560.00	9.30	114.83	558.85	-13.32	22.21	21.79	1.327	1.320	0.840
570.00	9.71	115.07	568.71	-14.02	23.71	23.41	1.236	1.230	0.720
580.00	10.10	115.21	578.56	-14.75	25.27	25.11	1.172	1.170	0.420
590.00	10.68	115.04	588.40	-15.52	26.90	26.89	1.742	1.740	-0.510
600.00	11.23	115.09	598.22	-16.32	28.62	28.76	1.650	1.650	0.150
610.00	11.89	115.65	608.01	-17.18	30.43	30.74	2.008	1.980	1.680
620.00	12.15	115.66	617.80	-18.08	32.31	32.80	0.780	0.780	0.030
630.00	12.39	115.86	627.57	-19.00	34.22	34.89	0.731	0.720	0.600
640.00	12.77	116.66	637.33	-19.97	36.17	37.05	1.254	1.140	2.400
650.00	13.11	117.53	647.07	-20.99	38.17	39.27	1.176	1.020	2.610
660.00	13.29	118.11	656.81	-22.05	40.19	41.54	0.670	0.540	1.740

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Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
670.00	13.46	118.65	666.54	-23.15	42.22	43.84	0.633	0.510	1.620
680.00	13.71	119.67	676.26	-24.30	44.27	46.17	1.039	0.750	3.060
690.00	13.91	121.43	685.97	-25.51	46.33	48.55	1.396	0.600	5.280
700.00	14.02	122.28	695.67	-26.79	48.38	50.96	0.698	0.330	2.550
710.00	14.19	123.08	705.37	-28.10	50.43	53.40	0.776	0.510	2.400
720.00	14.44	124.37	715.06	-29.47	52.49	55.87	1.216	0.750	3.870
730.00	14.75	125.72	724.74	-30.92	54.55	58.39	1.381	0.930	4.050
740.00	14.97	126.52	734.40	-32.43	56.62	60.95	0.902	0.660	2.400
750.00	15.31	127.07	744.06	-34.00	58.71	63.56	1.107	1.020	1.650
760.00	15.80	128.00	753.69	-35.63	60.84	66.24	1.649	1.470	2.790
770.00	16.34	128.60	763.30	-37.35	63.01	69.01	1.695	1.620	1.800
780.00	16.87	128.67	772.88	-39.13	65.24	71.86	1.591	1.590	0.210
790.00	17.50	128.79	782.44	-40.98	67.55	74.80	1.893	1.890	0.360
800.00	18.40	128.93	791.95	-42.91	69.95	77.88	2.703	2.700	0.420
810.00	19.13	128.69	801.42	-44.93	72.45	81.09	2.202	2.190	-0.720
820.00	19.61	128.36	810.85	-47.00	75.05	84.40	1.477	1.440	-0.990
830.00	20.09	128.11	820.26	-49.10	77.71	87.78	1.462	1.440	-0.750
840.00	20.61	127.67	829.63	-51.23	80.46	91.26	1.626	1.560	-1.320
850.00	21.04	126.88	838.98	-53.39	83.29	94.81	1.541	1.290	-2.370
860.00	21.40	126.14	848.30	-55.54	86.20	98.43	1.346	1.080	-2.220
870.00	21.66	125.36	857.60	-57.68	89.18	102.09	1.160	0.780	-2.340
880.00	21.71	124.64	866.90	-59.80	92.20	105.79	0.812	0.150	-2.160
890.00	21.70	124.31	876.19	-61.90	95.25	109.49	0.367	-0.030	-0.990
900.00	21.25	124.06	885.49	-63.95	98.28	113.15	1.378	-1.350	-0.750
910.00	20.45	123.59	894.84	-65.93	101.24	116.71	2.452	-2.400	-1.410
920.00	19.76	123.32	904.23	-67.83	104.10	120.14	2.089	-2.070	-0.810
930.00	19.35	123.24	913.65	-69.67	106.90	123.49	1.233	-1.230	-0.240
940.00	19.05	123.04	923.10	-71.46	109.66	126.78	0.921	-0.900	-0.600
950.00	18.82	122.42	932.56	-73.22	112.39	130.02	0.917	-0.690	-1.860
960.00	18.76	121.53	942.02	-74.92	115.12	133.24	0.879	-0.180	-2.670
970.00	18.76	120.39	951.49	-76.58	117.88	136.45	1.100	0.000	-3.420
980.00	18.83	119.53	960.96	-78.19	120.67	139.66	0.857	0.210	-2.580
990.00	18.87	118.52	970.42	-79.75	123.49	142.87	0.986	0.120	-3.030
1,000.00	18.94	117.69	979.88	-81.28	126.35	146.09	0.834	0.210	-2.490
1,010.00	18.99	117.08	989.34	-82.78	129.24	149.31	0.613	0.150	-1.830
1,020.00	19.11	116.31	998.79	-84.24	132.15	152.55	0.835	0.360	-2.310
1,030.00	19.38	115.18	1,008.23	-85.67	135.12	155.80	1.380	0.810	-3.390
1,040.00	19.81	113.76	1,017.65	-87.06	138.17	159.10	1.925	1.290	-4.260
1,050.00	20.19	113.30	1,027.05	-88.43	141.31	162.46	1.234	1.140	-1.380
1,060.00	20.45	113.25	1,036.43	-89.80	144.50	165.86	0.782	0.780	-0.150
1,070.00	20.56	113.30	1,045.79	-91.18	147.72	169.29	0.334	0.330	0.150
1,080.00	20.72	113.64	1,055.15	-92.59	150.95	172.75	0.600	0.480	1.020
1,090.00	20.96	114.82	1,064.50	-94.05	154.19	176.25	1.451	0.720	3.540

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Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,100.00	21.31	116.57	1,073.83	-95.61	157.44	179.81	2.165	1.050	5.250
1,110.00	21.80	119.11	1,083.13	-97.33	160.69	183.46	3.162	1.470	7.620
1,120.00	22.42	121.45	1,092.39	-99.23	163.94	187.21	3.231	1.860	7.020
1,130.00	22.93	122.76	1,101.62	-101.27	167.20	191.06	2.153	1.530	3.930
1,140.00	23.46	123.56	1,110.81	-103.43	170.50	195.00	1.850	1.590	2.400
1,150.00	24.23	123.79	1,119.96	-105.67	173.87	199.04	2.327	2.310	0.690
1,160.00	25.03	123.68	1,129.05	-107.99	177.33	203.21	2.404	2.400	-0.330
1,170.00	25.65	123.27	1,138.08	-110.35	180.90	207.49	1.933	1.860	-1.230
1,180.00	26.32	122.53	1,147.07	-112.73	184.58	211.87	2.233	2.010	-2.220
1,190.00	26.84	121.79	1,156.02	-115.11	188.37	216.34	1.849	1.560	-2.220
1,200.00	27.31	120.99	1,164.92	-117.48	192.25	220.88	1.784	1.410	-2.400
1,210.00	27.87	120.37	1,173.78	-119.84	196.24	225.50	1.888	1.680	-1.860
1,220.00	28.39	119.92	1,182.60	-122.21	200.31	230.20	1.685	1.560	-1.350
1,230.00	29.01	119.62	1,191.37	-124.59	204.48	234.98	1.910	1.860	-0.900
1,240.00	29.71	119.54	1,200.09	-127.01	208.75	239.87	2.103	2.100	-0.240
1,250.00	30.28	119.50	1,208.75	-129.48	213.10	244.85	1.711	1.710	-0.120
1,260.00	30.70	119.42	1,217.37	-131.97	217.51	249.90	1.266	1.260	-0.240
1,270.00	30.84	119.31	1,225.96	-134.48	221.97	254.99	0.453	0.420	-0.330
1,280.00	30.61	119.28	1,234.56	-136.98	226.43	260.08	0.692	-0.690	-0.090
1,290.00	30.14	119.39	1,243.18	-139.46	230.84	265.11	1.420	-1.410	0.330
1,300.00	29.48	119.45	1,251.86	-141.90	235.17	270.06	1.982	-1.980	0.180
1,310.00	29.02	119.52	1,260.58	-144.30	239.42	274.93	1.384	-1.380	0.210
1,320.00	28.79	119.74	1,269.34	-146.69	243.62	279.74	0.760	-0.690	0.660
1,330.00	28.51	119.93	1,278.11	-149.08	247.78	284.52	0.883	-0.840	0.570
1,340.00	27.89	120.05	1,286.93	-151.44	251.87	289.23	1.868	-1.860	0.360
1,350.00	27.26	120.01	1,295.79	-153.76	255.88	293.84	1.891	-1.890	-0.120
1,360.00	27.20	119.88	1,304.68	-156.04	259.85	298.40	0.253	-0.180	-0.390
1,370.00	27.23	119.87	1,313.58	-158.32	263.81	302.96	0.091	0.090	-0.030
1,380.00	27.38	119.62	1,322.46	-160.60	267.79	307.53	0.566	0.450	-0.750
1,390.00	27.64	119.39	1,331.33	-162.87	271.81	312.13	0.843	0.780	-0.690
1,400.00	27.93	119.42	1,340.18	-165.16	275.88	316.77	0.871	0.870	0.090
1,410.00	28.16	119.41	1,349.00	-167.47	279.97	321.45	0.690	0.690	-0.030
1,420.00	28.21	119.23	1,357.82	-169.78	284.09	326.16	0.296	0.150	-0.540
1,430.00	27.99	119.18	1,366.64	-172.08	288.20	330.85	0.664	-0.660	-0.150
1,440.00	27.72	119.17	1,375.48	-174.36	292.28	335.50	0.810	-0.810	-0.030
1,450.00	27.54	119.15	1,384.34	-176.62	296.33	340.11	0.541	-0.540	-0.060
1,460.00	27.36	119.06	1,393.21	-178.86	300.36	344.70	0.554	-0.540	-0.270
1,470.00	27.29	118.91	1,402.10	-181.08	304.37	349.27	0.295	-0.210	-0.450
1,480.00	27.20	118.89	1,410.99	-183.30	308.38	353.82	0.271	-0.270	-0.060
1,490.00	27.05	118.75	1,419.89	-185.49	312.38	358.36	0.489	-0.450	-0.420
1,500.00	26.93	118.56	1,428.80	-187.67	316.36	362.87	0.443	-0.360	-0.570
1,510.00	26.85	118.33	1,437.72	-189.82	320.34	367.37	0.394	-0.240	-0.690
1,520.00	26.81	118.13	1,446.64	-191.96	324.31	371.85	0.296	-0.120	-0.600
1,530.00	26.64	118.00	1,455.57	-194.07	328.28	376.32	0.539	-0.510	-0.390

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Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,540.00	26.54	117.85	1,464.52	-196.17	332.23	380.76	0.361	-0.300	-0.450
1,550.00	26.52	117.68	1,473.46	-198.25	336.19	385.20	0.236	-0.060	-0.510
1,560.00	26.44	117.60	1,482.41	-200.32	340.14	389.62	0.263	-0.240	-0.240
1,570.00	26.29	117.54	1,491.37	-202.38	344.07	394.03	0.457	-0.450	-0.180
1,580.00	26.20	117.39	1,500.34	-204.42	348.00	398.41	0.335	-0.270	-0.450
1,590.00	26.13	117.30	1,509.32	-206.44	351.91	402.79	0.241	-0.210	-0.270
1,600.00	26.06	117.17	1,518.30	-208.45	355.83	407.15	0.271	-0.210	-0.390
1,610.00	25.95	117.06	1,527.29	-210.45	359.73	411.50	0.360	-0.330	-0.330
1,620.00	25.84	116.89	1,536.28	-212.43	363.62	415.82	0.398	-0.330	-0.510
1,630.00	25.78	116.67	1,545.29	-214.39	367.51	420.14	0.339	-0.180	-0.660
1,640.00	25.69	116.57	1,554.29	-216.34	371.39	424.43	0.300	-0.270	-0.300
1,650.00	25.56	116.39	1,563.31	-218.27	375.26	428.71	0.455	-0.390	-0.540
1,660.00	25.44	116.21	1,572.34	-220.18	379.12	432.97	0.429	-0.360	-0.540
1,670.00	25.43	116.11	1,581.37	-222.07	382.97	437.22	0.132	-0.030	-0.300
1,680.00	25.43	116.03	1,590.40	-223.96	386.83	441.47	0.103	0.000	-0.240
1,690.00	25.53	116.10	1,599.43	-225.85	390.70	445.72	0.313	0.300	0.210
1,700.00	25.86	116.06	1,608.44	-227.75	394.59	450.01	0.991	0.990	-0.120
1,710.00	26.26	116.26	1,617.42	-229.69	398.53	454.35	1.229	1.200	0.600
1,720.00	26.89	116.56	1,626.36	-231.68	402.54	458.78	1.932	1.890	0.900
1,730.00	27.50	116.94	1,635.26	-233.74	406.62	463.30	1.903	1.830	1.140
1,740.00	28.01	117.15	1,644.11	-235.85	410.77	467.92	1.558	1.530	0.630
1,750.00	28.50	117.23	1,652.92	-238.02	414.98	472.61	1.474	1.470	0.240
1,760.00	28.97	117.56	1,661.68	-240.23	419.25	477.38	1.488	1.410	0.990
1,770.00	29.41	118.00	1,670.41	-242.50	423.56	482.22	1.469	1.320	1.320
1,780.00	29.85	118.48	1,679.11	-244.84	427.92	487.14	1.500	1.320	1.440
1,790.00	30.32	118.84	1,687.76	-247.25	432.32	492.12	1.510	1.410	1.080
1,800.00	30.70	119.23	1,696.37	-249.71	436.76	497.17	1.285	1.140	1.170
1,810.00	31.13	119.68	1,704.95	-252.24	441.23	502.29	1.465	1.290	1.350
1,820.00	31.61	119.95	1,713.49	-254.83	445.75	507.48	1.500	1.440	0.810
1,830.00	32.07	120.45	1,721.99	-257.48	450.30	512.73	1.591	1.380	1.500
1,840.00	32.58	121.01	1,730.44	-260.21	454.90	518.07	1.774	1.530	1.680
1,850.00	33.17	121.55	1,738.84	-263.03	459.54	523.49	1.976	1.770	1.620
1,860.00	33.78	122.26	1,747.18	-265.95	464.22	529.00	2.175	1.830	2.130
1,870.00	34.41	122.96	1,755.46	-268.97	468.94	534.60	2.227	1.890	2.100
1,880.00	35.08	123.78	1,763.68	-272.10	473.70	540.30	2.451	2.010	2.460
1,890.00	35.74	124.55	1,771.83	-275.36	478.50	546.09	2.390	1.980	2.310
1,900.00	36.39	124.97	1,779.91	-278.71	483.33	551.98	2.086	1.950	1.260
1,910.00	37.11	125.42	1,787.92	-282.16	488.22	557.96	2.306	2.160	1.350
1,920.00	37.85	125.91	1,795.86	-285.71	493.17	564.04	2.393	2.220	1.470
1,930.00	38.51	126.34	1,803.72	-289.35	498.16	570.22	2.135	1.980	1.290
1,940.00	39.27	126.74	1,811.50	-293.09	503.20	576.50	2.401	2.280	1.200
1,950.00	40.09	127.17	1,819.20	-296.93	508.31	582.88	2.594	2.460	1.290
1,960.00	40.87	127.46	1,826.80	-300.87	513.47	589.36	2.407	2.340	0.870

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,970.00	41.72	127.60	1,834.32	-304.89	518.70	595.95	2.565	2.550	0.420
1,980.00	42.38	127.57	1,841.74	-308.97	524.01	602.64	1.981	1.980	-0.090
1,990.00	43.08	127.46	1,849.09	-313.10	529.39	609.42	2.112	2.100	-0.330
2,000.00	43.74	127.26	1,856.35	-317.27	534.85	616.28	2.022	1.980	-0.600
2,010.00	44.38	127.07	1,863.54	-321.47	540.40	623.23	1.960	1.920	-0.570
2,020.00	44.97	126.80	1,870.65	-325.70	546.02	630.26	1.859	1.770	-0.810
2,030.00	45.09	126.48	1,877.72	-329.92	551.69	637.33	0.769	0.360	-0.960
2,040.00	44.90	126.27	1,884.79	-334.11	557.38	644.40	0.723	-0.570	-0.630
2,050.00	44.83	126.20	1,891.88	-338.28	563.07	651.45	0.257	-0.210	-0.210
2,060.00	44.96	126.17	1,898.96	-342.45	568.77	658.50	0.395	0.390	-0.090
2,070.00	44.98	126.04	1,906.04	-346.62	574.48	665.57	0.282	0.060	-0.390
2,080.00	44.96	125.96	1,913.11	-350.77	580.20	672.63	0.180	-0.060	-0.240
2,090.00	44.99	125.95	1,920.18	-354.92	585.92	679.70	0.092	0.090	-0.030
2,100.00	45.05	125.87	1,927.25	-359.07	591.65	686.77	0.247	0.180	-0.240
2,110.00	45.04	125.70	1,934.32	-363.21	597.39	693.85	0.362	-0.030	-0.510
2,120.00	45.00	125.59	1,941.39	-367.33	603.14	700.92	0.262	-0.120	-0.330
2,130.00	45.05	125.55	1,948.46	-371.44	608.89	707.99	0.172	0.150	-0.120
2,140.00	45.17	125.51	1,955.51	-375.56	614.66	715.08	0.370	0.360	-0.120
2,150.00	45.15	125.42	1,962.56	-379.67	620.43	722.17	0.201	-0.060	-0.270
2,160.00	44.93	125.42	1,969.63	-383.78	626.20	729.24	0.660	-0.660	0.000
2,170.00	44.95	125.48	1,976.71	-387.87	631.95	736.31	0.141	0.060	0.180
2,180.00	45.03	125.45	1,983.78	-391.97	637.71	743.38	0.248	0.240	-0.090
2,190.00	44.99	125.31	1,990.85	-396.07	643.48	750.45	0.320	-0.120	-0.420
2,200.00	44.96	125.21	1,997.92	-400.15	649.25	757.52	0.230	-0.090	-0.300
2,210.00	45.07	125.06	2,004.99	-404.22	655.03	764.59	0.458	0.330	-0.450
2,220.00	45.12	124.82	2,012.05	-408.28	660.84	771.67	0.532	0.150	-0.720
2,230.00	45.17	124.56	2,019.11	-412.31	666.67	778.76	0.573	0.150	-0.780
2,240.00	45.31	124.29	2,026.15	-416.33	672.53	785.86	0.712	0.420	-0.810
2,250.00	45.35	124.14	2,033.18	-420.32	678.41	792.97	0.342	0.120	-0.450
2,260.00	45.14	124.16	2,040.22	-424.31	684.28	800.07	0.631	-0.630	0.060
2,270.00	44.93	124.15	2,047.29	-428.28	690.14	807.15	0.630	-0.630	-0.030
2,280.00	44.92	124.20	2,054.37	-432.25	695.98	814.21	0.110	-0.030	0.150
2,290.00	44.98	124.16	2,061.44	-436.22	701.83	821.27	0.199	0.180	-0.120
2,300.00	44.91	124.07	2,068.52	-440.18	707.67	828.34	0.284	-0.210	-0.270
2,310.00	45.14	124.09	2,075.59	-444.14	713.53	835.41	0.691	0.690	0.060
2,320.00	45.51	124.06	2,082.62	-448.13	719.42	842.52	1.112	1.110	-0.090
2,330.00	45.53	124.14	2,089.63	-452.13	725.33	849.66	0.181	0.060	0.240
2,340.00	45.15	124.04	2,096.66	-456.12	731.22	856.77	1.160	-1.140	-0.300
2,350.00	44.84	124.02	2,103.73	-460.07	737.08	863.84	0.931	-0.930	-0.060
2,360.00	45.06	124.05	2,110.80	-464.03	742.94	870.90	0.663	0.660	0.090
2,370.00	45.16	124.09	2,117.86	-468.00	748.81	877.99	0.312	0.300	0.120
2,380.00	44.89	124.15	2,124.93	-471.96	754.66	885.06	0.820	-0.810	0.180
2,390.00	44.97	124.28	2,132.01	-475.94	760.50	892.12	0.365	0.240	0.390

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,400.00	45.21	124.34	2,139.07	-479.93	766.35	899.21	0.731	0.720	0.180	
2,410.00	45.38	124.58	2,146.10	-483.95	772.21	906.31	0.722	0.510	0.720	
2,420.00	45.14	124.85	2,153.14	-487.99	778.05	913.42	0.922	-0.720	0.810	
2,430.00	45.02	125.03	2,160.20	-492.05	783.86	920.50	0.525	-0.360	0.540	
2,440.00	45.17	125.07	2,167.26	-496.12	789.65	927.58	0.458	0.450	0.120	
2,450.00	45.03	125.06	2,174.32	-500.19	795.45	934.66	0.421	-0.420	-0.030	
2,460.00	44.94	124.92	2,181.40	-504.24	801.24	941.73	0.401	-0.270	-0.420	
2,470.00	44.92	124.87	2,188.47	-508.28	807.04	948.79	0.122	-0.060	-0.150	
2,480.00	44.93	124.99	2,195.56	-512.32	812.82	955.86	0.256	0.030	0.360	
2,490.00	44.84	124.95	2,202.64	-516.37	818.61	962.91	0.283	-0.270	-0.120	
2,500.00	44.80	125.09	2,209.73	-520.41	824.38	969.96	0.319	-0.120	0.420	
2,510.00	44.68	125.10	2,216.84	-524.46	830.14	977.00	0.361	-0.360	0.030	
2,520.00	44.49	125.22	2,223.96	-528.50	835.88	984.02	0.624	-0.570	0.360	
2,530.00	44.67	125.19	2,231.08	-532.55	841.61	991.04	0.544	0.540	-0.090	
2,540.00	44.71	125.34	2,238.19	-536.61	847.36	998.07	0.338	0.120	0.450	
2,550.00	44.65	125.31	2,245.30	-540.68	853.09	1,005.10	0.191	-0.180	-0.090	
2,560.00	44.75	125.45	2,252.41	-544.75	858.83	1,012.14	0.421	0.300	0.420	
2,570.00	44.84	125.62	2,259.51	-548.84	864.56	1,019.18	0.449	0.270	0.510	
2,580.00	44.87	125.68	2,266.59	-552.95	870.29	1,026.23	0.156	0.090	0.180	
2,590.00	44.90	125.79	2,273.68	-557.07	876.02	1,033.29	0.250	0.090	0.330	
2,600.00	44.72	125.88	2,280.77	-561.20	881.73	1,040.33	0.573	-0.540	0.270	
2,610.00	44.39	125.99	2,287.90	-565.32	887.41	1,047.35	1.017	-0.990	0.330	
2,620.00	44.49	125.95	2,295.04	-569.43	893.08	1,054.35	0.312	0.300	-0.120	
2,630.00	44.76	125.94	2,302.16	-573.55	898.77	1,061.37	0.810	0.810	-0.030	
2,640.00	44.86	125.88	2,309.25	-577.69	904.48	1,068.42	0.326	0.300	-0.180	
2,650.00	45.01	125.81	2,316.33	-581.82	910.20	1,075.48	0.474	0.450	-0.210	
2,660.00	45.09	125.87	2,323.40	-585.97	915.94	1,082.55	0.272	0.240	0.180	
2,670.00	45.15	125.98	2,330.45	-590.13	921.68	1,089.64	0.295	0.180	0.330	
2,680.00	45.27	125.99	2,337.50	-594.30	927.42	1,096.73	0.361	0.360	0.030	
2,690.00	45.33	126.02	2,344.53	-598.47	933.17	1,103.84	0.191	0.180	0.090	
2,700.00	45.54	126.20	2,351.55	-602.67	938.93	1,110.96	0.738	0.630	0.540	
2,710.00	45.74	126.17	2,358.54	-606.89	944.70	1,118.11	0.603	0.600	-0.090	
2,720.00	45.86	126.18	2,365.51	-611.13	950.48	1,125.28	0.361	0.360	0.030	
2,730.00	45.90	126.29	2,372.47	-615.37	956.27	1,132.45	0.266	0.120	0.330	
2,740.00	46.03	126.35	2,379.42	-619.63	962.07	1,139.64	0.411	0.390	0.180	
2,750.00	46.22	126.40	2,386.35	-623.90	967.87	1,146.84	0.580	0.570	0.150	
2,760.00	46.26	126.46	2,393.27	-628.19	973.68	1,154.06	0.177	0.120	0.180	
2,770.00	46.32	126.64	2,400.18	-632.50	979.49	1,161.29	0.430	0.180	0.540	
2,780.00	46.52	126.73	2,407.07	-636.82	985.30	1,168.53	0.631	0.600	0.270	
2,790.00	46.83	126.63	2,413.94	-641.17	991.13	1,175.80	0.955	0.930	-0.300	
2,800.00	47.00	126.73	2,420.77	-645.53	996.99	1,183.10	0.555	0.510	0.300	
2,810.00	46.99	126.76	2,427.59	-649.91	1,002.85	1,190.41	0.072	-0.030	0.090	
2,820.00	47.38	126.81	2,434.38	-654.30	1,008.72	1,197.74	1.175	1.170	0.150	
2,830.00	48.02	126.64	2,441.11	-658.72	1,014.65	1,205.13	1.957	1.920	-0.510	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,840.00	49.07	126.33	2,447.73	-663.18	1,020.68	1,212.62	3.226	3.150	-0.930
2,850.00	50.14	126.22	2,454.21	-667.68	1,026.82	1,220.23	3.220	3.210	-0.330
2,860.00	50.83	126.14	2,460.58	-672.24	1,033.04	1,227.94	2.078	2.070	-0.240
2,870.00	51.01	125.98	2,466.88	-676.81	1,039.32	1,235.70	0.656	0.540	-0.480
2,880.00	51.24	125.73	2,473.16	-681.37	1,045.63	1,243.49	0.904	0.690	-0.750
2,890.00	52.02	125.36	2,479.37	-685.93	1,052.01	1,251.33	2.497	2.340	-1.110
2,900.00	53.05	125.00	2,485.45	-690.50	1,058.50	1,259.26	3.207	3.090	-1.080
2,910.00	53.79	124.82	2,491.41	-695.09	1,065.08	1,267.29	2.262	2.220	-0.540
2,920.00	54.47	124.63	2,497.27	-699.71	1,071.74	1,275.40	2.092	2.040	-0.570
2,930.00	55.24	124.38	2,503.02	-704.34	1,078.48	1,283.57	2.390	2.310	-0.750
2,940.00	56.18	124.21	2,508.66	-709.00	1,085.31	1,291.84	2.851	2.820	-0.510
2,950.00	56.90	123.93	2,514.17	-713.67	1,092.22	1,300.18	2.271	2.160	-0.840
2,960.00	57.46	123.87	2,519.59	-718.36	1,099.19	1,308.58	1.687	1.680	-0.180
2,970.00	58.29	123.56	2,524.91	-723.06	1,106.24	1,317.05	2.612	2.490	-0.930
2,980.00	59.29	123.30	2,530.09	-727.77	1,113.37	1,325.60	3.073	3.000	-0.780
2,990.00	60.01	123.02	2,535.14	-732.49	1,120.60	1,334.23	2.278	2.160	-0.840
3,000.00	60.49	122.74	2,540.10	-737.20	1,127.89	1,342.90	1.614	1.440	-0.840
3,010.00	61.29	122.44	2,544.97	-741.91	1,135.25	1,351.63	2.526	2.400	-0.900
3,020.00	62.30	121.86	2,549.70	-746.60	1,142.71	1,360.44	3.396	3.030	-1.740
3,030.00	62.91	121.69	2,554.30	-751.27	1,150.26	1,369.30	1.885	1.830	-0.510
3,040.00	63.59	121.65	2,558.80	-755.96	1,157.86	1,378.22	2.043	2.040	-0.120
3,050.00	64.51	121.28	2,563.17	-760.65	1,165.53	1,387.20	2.935	2.760	-1.110
3,060.00	65.51	121.14	2,567.40	-765.35	1,173.28	1,396.25	3.024	3.000	-0.420
3,070.00	66.18	121.02	2,571.49	-770.06	1,181.10	1,405.35	2.037	2.010	-0.360
3,080.00	66.79	120.81	2,575.48	-774.77	1,188.96	1,414.50	1.919	1.830	-0.630
3,090.00	67.90	120.52	2,579.33	-779.48	1,196.90	1,423.71	3.425	3.330	-0.870
3,100.00	69.07	120.34	2,583.00	-784.19	1,204.92	1,432.98	3.546	3.510	-0.540
3,110.00	69.64	120.09	2,586.53	-788.90	1,213.01	1,442.31	1.848	1.710	-0.750
3,120.00	70.42	120.01	2,589.94	-793.61	1,221.14	1,451.68	2.351	2.340	-0.240
3,130.00	71.46	120.03	2,593.21	-798.34	1,229.33	1,461.10	3.121	3.120	0.060
3,140.00	71.83	120.16	2,596.35	-803.09	1,237.54	1,470.56	1.170	1.110	0.390
3,150.00	72.15	119.95	2,599.45	-807.86	1,245.77	1,480.04	1.132	0.960	-0.630
3,160.00	72.75	119.90	2,602.46	-812.61	1,254.03	1,489.54	1.806	1.800	-0.150
3,170.00	73.28	119.86	2,605.38	-817.38	1,262.33	1,499.07	1.594	1.590	-0.120
3,180.00	73.45	119.87	2,608.25	-822.15	1,270.64	1,508.62	0.511	0.510	0.030
3,190.00	73.60	119.88	2,611.08	-826.93	1,278.95	1,518.18	0.451	0.450	0.030
3,200.00	74.64	119.90	2,613.82	-831.72	1,287.29	1,527.76	3.121	3.120	0.060
3,210.00	75.28	119.90	2,616.41	-836.53	1,295.66	1,537.38	1.920	1.920	0.000
3,220.00	75.66	120.01	2,618.92	-841.37	1,304.05	1,547.03	1.184	1.140	0.330
3,230.00	76.61	120.15	2,621.32	-846.23	1,312.45	1,556.71	2.879	2.850	0.420
3,240.00	77.52	119.97	2,623.56	-851.11	1,320.88	1,566.42	2.780	2.730	-0.540
3,250.00	78.06	119.96	2,625.67	-856.00	1,329.35	1,576.16	1.620	1.620	-0.030
3,260.00	78.27	119.88	2,627.72	-860.88	1,337.84	1,585.91	0.672	0.630	-0.240

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3,270.00	78.47	119.80	2,629.74	-865.75	1,346.33	1,595.67	0.644	0.600	-0.240
3,280.00	79.66	119.87	2,631.64	-870.64	1,354.85	1,605.46	3.576	3.570	0.210
3,290.00	80.27	119.95	2,633.38	-875.55	1,363.38	1,615.27	1.845	1.830	0.240
3,300.00	80.48	120.06	2,635.05	-880.48	1,371.92	1,625.10	0.709	0.630	0.330
3,310.00	80.87	120.05	2,636.67	-885.42	1,380.46	1,634.93	1.170	1.170	-0.030
3,320.00	82.06	119.97	2,638.15	-890.37	1,389.03	1,644.79	3.578	3.570	-0.240
3,330.00	82.23	120.40	2,639.52	-895.35	1,397.59	1,654.66	1.376	0.510	1.290
3,340.00	82.87	120.69	2,640.82	-900.39	1,406.13	1,664.55	2.105	1.920	0.870
3,350.00	83.39	120.89	2,642.01	-905.47	1,414.66	1,674.46	1.670	1.560	0.600
3,360.00	84.45	120.87	2,643.07	-910.57	1,423.19	1,684.38	3.181	3.180	-0.060
3,370.00	84.59	120.93	2,644.03	-915.68	1,431.73	1,694.31	0.457	0.420	0.180
3,380.00	84.95	121.06	2,644.94	-920.81	1,440.27	1,704.25	1.148	1.080	0.390
3,390.00	85.99	121.01	2,645.73	-925.95	1,448.81	1,714.20	3.124	3.120	-0.150
3,400.00	85.90	121.04	2,646.44	-931.09	1,457.36	1,724.15	0.285	-0.270	0.090
3,410.00	85.60	121.10	2,647.18	-936.24	1,465.90	1,734.10	0.918	-0.900	0.180
3,420.00	85.55	121.37	2,647.95	-941.41	1,474.43	1,744.06	0.821	-0.150	0.810
3,430.00	86.03	121.83	2,648.68	-946.63	1,482.92	1,754.01	1.992	1.440	1.380
3,440.00	85.95	121.95	2,649.38	-951.90	1,491.39	1,763.98	0.432	-0.240	0.360
3,450.00	86.13	122.07	2,650.07	-957.19	1,499.85	1,773.94	0.649	0.540	0.360
3,460.00	86.45	121.97	2,650.72	-962.48	1,508.31	1,783.91	1.006	0.960	-0.300
3,470.00	86.18	121.91	2,651.36	-967.76	1,516.78	1,793.88	0.830	-0.810	-0.180
3,480.00	85.93	121.92	2,652.05	-973.04	1,525.25	1,803.84	0.751	-0.750	0.030
3,490.00	85.80	122.49	2,652.77	-978.35	1,533.69	1,813.81	1.750	-0.390	1.710
3,500.00	85.83	122.79	2,653.50	-983.73	1,542.08	1,823.77	0.902	0.090	0.900
3,510.00	86.12	122.81	2,654.20	-989.13	1,550.47	1,833.74	0.872	0.870	0.060
3,520.00	85.84	122.86	2,654.90	-994.54	1,558.85	1,843.71	0.853	-0.840	0.150
3,530.00	85.78	122.90	2,655.64	-999.96	1,567.23	1,853.68	0.216	-0.180	0.120
3,540.00	86.22	123.03	2,656.33	-1,005.39	1,575.60	1,863.65	1.376	1.320	0.390
3,550.00	86.14	123.04	2,657.00	-1,010.83	1,583.96	1,873.63	0.242	-0.240	0.030
3,560.00	85.83	123.11	2,657.70	-1,016.27	1,592.32	1,883.60	0.953	-0.930	0.210
3,570.00	85.96	123.13	2,658.42	-1,021.72	1,600.67	1,893.57	0.395	0.390	0.060
3,580.00	86.23	123.23	2,659.10	-1,027.18	1,609.02	1,903.54	0.864	0.810	0.300
3,590.00	86.12	123.25	2,659.76	-1,032.65	1,617.37	1,913.52	0.335	-0.330	0.060
3,600.00	86.14	123.27	2,660.44	-1,038.12	1,625.71	1,923.49	0.085	0.060	0.060
3,610.00	86.03	123.23	2,661.12	-1,043.59	1,634.06	1,933.46	0.351	-0.330	-0.120
3,620.00	86.10	123.08	2,661.81	-1,049.05	1,642.41	1,943.44	0.496	0.210	-0.450
3,630.00	86.18	123.04	2,662.48	-1,054.49	1,650.77	1,953.41	0.268	0.240	-0.120
3,640.00	85.99	122.98	2,663.16	-1,059.92	1,659.14	1,963.38	0.598	-0.570	-0.180
3,650.00	86.00	122.95	2,663.86	-1,065.35	1,667.50	1,973.35	0.095	0.030	-0.090
3,660.00	86.32	122.80	2,664.53	-1,070.77	1,675.88	1,983.33	1.060	0.960	-0.450
3,670.00	85.85	122.53	2,665.21	-1,076.15	1,684.28	1,993.30	1.625	-1.410	-0.810
3,680.00	85.76	121.87	2,665.95	-1,081.47	1,692.72	2,003.26	1.993	-0.270	-1.980
3,690.00	86.02	121.35	2,666.66	-1,086.70	1,701.22	2,013.22	1.741	0.780	-1.560

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3,700.00	86.30	120.85	2,667.33	-1,091.85	1,709.76	2,023.18	1.716	0.840	-1.500
3,710.00	86.05	120.29	2,668.00	-1,096.92	1,718.35	2,033.13	1.836	-0.750	-1.680
3,720.00	86.23	119.34	2,668.67	-1,101.88	1,727.01	2,043.07	2.894	0.540	-2.850
3,730.00	86.70	118.89	2,669.29	-1,106.74	1,735.73	2,053.00	1.950	1.410	-1.350
3,740.00	86.35	118.65	2,669.90	-1,111.54	1,744.48	2,062.93	1.272	-1.050	-0.720
3,750.00	85.93	118.53	2,670.57	-1,116.32	1,753.24	2,072.85	1.310	-1.260	-0.360
3,760.00	85.48	118.59	2,671.32	-1,121.09	1,762.00	2,082.77	1.362	-1.350	0.180
3,770.00	85.93	119.00	2,672.07	-1,125.89	1,770.74	2,092.69	1.824	1.350	1.230
3,780.00	85.89	119.61	2,672.78	-1,130.77	1,779.43	2,102.62	1.829	-0.120	1.830
3,790.00	85.55	120.05	2,673.53	-1,135.73	1,788.08	2,112.55	1.665	-1.020	1.320
3,800.00	85.75	120.10	2,674.28	-1,140.73	1,796.71	2,122.49	0.618	0.600	0.150
3,810.00	85.80	120.30	2,675.02	-1,145.75	1,805.33	2,132.43	0.617	0.150	0.600
3,820.00	85.46	120.45	2,675.78	-1,150.79	1,813.93	2,142.38	1.114	-1.020	0.450
3,830.00	85.24	120.94	2,676.59	-1,155.88	1,822.50	2,152.32	1.607	-0.660	1.470
3,840.00	85.58	121.76	2,677.39	-1,161.06	1,831.02	2,162.27	2.656	1.020	2.460
3,850.00	85.72	122.47	2,678.15	-1,166.36	1,839.46	2,172.23	2.165	0.420	2.130
3,860.00	85.31	122.80	2,678.93	-1,171.74	1,847.86	2,182.19	1.577	-1.230	0.990
3,870.00	85.39	123.16	2,679.74	-1,177.16	1,856.22	2,192.16	1.103	0.240	1.080
3,880.00	85.88	123.52	2,680.51	-1,182.64	1,864.55	2,202.12	1.822	1.470	1.080
3,890.00	85.86	123.59	2,681.23	-1,188.16	1,872.86	2,212.10	0.218	-0.060	0.210
3,900.00	85.92	123.62	2,681.94	-1,193.68	1,881.17	2,222.07	0.201	0.180	0.090
3,910.00	85.81	123.79	2,682.66	-1,199.21	1,889.47	2,232.04	0.606	-0.330	0.510
3,920.00	85.95	123.90	2,683.38	-1,204.77	1,897.75	2,242.01	0.534	0.420	0.330
3,930.00	85.73	123.96	2,684.11	-1,210.33	1,906.03	2,251.99	0.684	-0.660	0.180
3,940.00	85.71	124.07	2,684.85	-1,215.91	1,914.29	2,261.96	0.335	-0.060	0.330
3,950.00	85.71	124.27	2,685.60	-1,221.51	1,922.54	2,271.93	0.598	0.000	0.600
3,960.00	85.70	124.43	2,686.35	-1,227.14	1,930.77	2,281.90	0.480	-0.030	0.480
3,970.00	85.38	124.59	2,687.13	-1,232.79	1,938.99	2,291.87	1.073	-0.960	0.480
3,980.00	85.32	124.93	2,687.94	-1,238.47	1,947.18	2,301.84	1.032	-0.180	1.020
3,990.00	85.83	125.21	2,688.71	-1,244.20	1,955.34	2,311.81	1.744	1.530	0.840
4,000.00	86.07	125.44	2,689.42	-1,249.97	1,963.48	2,321.78	0.996	0.720	0.690
4,010.00	85.64	125.34	2,690.14	-1,255.74	1,971.61	2,331.76	1.324	-1.290	-0.300
4,020.00	85.44	125.40	2,690.92	-1,261.52	1,979.74	2,341.73	0.626	-0.600	0.180
4,030.00	85.69	125.55	2,691.69	-1,267.30	1,987.86	2,351.69	0.874	0.750	0.450
4,040.00	85.41	125.84	2,692.47	-1,273.12	1,995.95	2,361.66	1.207	-0.840	0.870
4,050.00	85.24	126.40	2,693.28	-1,278.99	2,004.00	2,371.63	1.750	-0.510	1.680
4,060.00	85.53	126.95	2,694.09	-1,284.95	2,012.00	2,381.59	1.861	0.870	1.650
4,070.00	86.26	127.34	2,694.80	-1,290.97	2,019.95	2,391.55	2.482	2.190	1.170
4,080.00	86.62	127.03	2,695.42	-1,297.00	2,027.90	2,401.52	1.424	1.080	-0.930
4,090.00	86.34	124.90	2,696.04	-1,302.86	2,035.98	2,411.50	6.433	-0.840	-6.390
4,100.00	86.47	123.07	2,696.66	-1,308.44	2,044.25	2,421.48	5.493	0.390	-5.490
4,110.00	86.49	121.36	2,697.28	-1,313.76	2,052.70	2,431.45	5.121	0.060	-5.130
4,120.00	86.26	119.40	2,697.91	-1,318.81	2,061.31	2,441.40	5.909	-0.690	-5.880
4,130.00	86.04	117.42	2,698.58	-1,323.56	2,070.08	2,451.32	5.963	-0.660	-5.940

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
4,140.00	86.42	115.75	2,699.24	-1,328.02	2,079.01	2,461.20	5.127	1.140	-5.010
4,150.00	86.34	114.12	2,699.87	-1,332.23	2,088.06	2,471.03	4.886	-0.240	-4.890
4,160.00	85.94	113.08	2,700.54	-1,336.22	2,097.20	2,480.82	3.336	-1.200	-3.120
4,170.00	85.39	113.19	2,701.30	-1,340.14	2,106.37	2,490.59	1.682	-1.650	0.330
4,180.00	85.31	113.86	2,702.11	-1,344.12	2,115.51	2,500.37	2.018	-0.240	2.010
4,190.00	85.27	115.00	2,702.93	-1,348.24	2,124.58	2,510.18	3.411	-0.120	3.420
4,200.00	85.50	116.66	2,703.74	-1,352.58	2,133.55	2,520.03	5.012	0.690	4.980
4,210.00	85.83	118.40	2,704.49	-1,357.19	2,142.39	2,529.92	5.298	0.990	5.220
4,220.00	86.06	119.43	2,705.20	-1,362.02	2,151.12	2,539.84	3.159	0.690	3.090
4,230.00	85.75	119.82	2,705.91	-1,366.95	2,159.79	2,549.78	1.492	-0.930	1.170
4,240.00	85.67	120.80	2,706.66	-1,371.98	2,168.40	2,559.72	2.942	-0.240	2.940
4,250.00	85.75	122.23	2,707.41	-1,377.19	2,176.90	2,569.68	4.285	0.240	4.290
4,260.00	85.46	123.61	2,708.18	-1,382.61	2,185.27	2,579.64	4.219	-0.870	4.140
4,270.00	85.41	124.95	2,708.97	-1,388.22	2,193.51	2,589.61	4.010	-0.150	4.020
4,280.00	85.42	126.33	2,709.77	-1,394.03	2,201.61	2,599.58	4.127	0.030	4.140
4,290.00	85.86	127.54	2,710.53	-1,400.02	2,209.58	2,609.54	3.853	1.320	3.630
4,300.00	85.95	128.45	2,711.25	-1,406.16	2,217.44	2,619.50	2.736	0.270	2.730
4,310.00	85.62	128.94	2,711.98	-1,412.40	2,225.23	2,629.45	1.769	-0.990	1.470
4,320.00	86.08	129.91	2,712.71	-1,418.73	2,232.93	2,639.39	3.214	1.380	2.910
4,330.00	86.40	130.43	2,713.36	-1,425.17	2,240.55	2,649.32	1.829	0.960	1.560
4,340.00	86.13	130.52	2,714.01	-1,431.65	2,248.15	2,659.25	0.854	-0.810	0.270
4,350.00	85.89	130.60	2,714.71	-1,438.13	2,255.72	2,669.17	0.759	-0.720	0.240
4,360.00	85.98	130.62	2,715.42	-1,444.63	2,263.30	2,679.09	0.277	0.270	0.060
4,370.00	86.21	130.61	2,716.10	-1,451.12	2,270.87	2,689.02	0.691	0.690	-0.030
4,380.00	86.01	130.64	2,716.78	-1,457.62	2,278.44	2,698.94	0.607	-0.600	0.090
4,390.00	85.70	130.53	2,717.50	-1,464.11	2,286.02	2,708.86	0.987	-0.930	-0.330
4,400.00	85.96	130.45	2,718.23	-1,470.58	2,293.60	2,718.78	0.816	0.780	-0.240
4,410.00	85.99	130.51	2,718.93	-1,477.06	2,301.19	2,728.71	0.201	0.090	0.180
4,420.00	85.90	130.64	2,719.64	-1,483.55	2,308.77	2,738.63	0.474	-0.270	0.390
4,430.00	85.97	130.60	2,720.34	-1,490.04	2,316.34	2,748.55	0.242	0.210	-0.120
4,440.00	86.13	130.48	2,721.03	-1,496.52	2,323.92	2,758.48	0.599	0.480	-0.360
4,450.00	86.07	130.53	2,721.71	-1,503.00	2,331.50	2,768.40	0.234	-0.180	0.150
4,460.00	86.09	130.64	2,722.40	-1,509.49	2,339.08	2,778.33	0.335	0.060	0.330
4,470.00	86.06	130.65	2,723.08	-1,515.99	2,346.65	2,788.25	0.095	-0.090	0.030
4,480.00	86.24	130.66	2,723.75	-1,522.49	2,354.22	2,798.17	0.541	0.540	0.030
4,490.00	86.31	130.86	2,724.40	-1,529.01	2,361.78	2,808.10	0.634	0.210	0.600
4,500.00	86.86	131.00	2,725.00	-1,535.55	2,369.32	2,818.02	1.702	1.650	0.420
4,510.00	86.82	130.92	2,725.55	-1,542.09	2,376.86	2,827.95	0.268	-0.120	-0.240
4,520.00	87.53	131.05	2,726.04	-1,548.64	2,384.40	2,837.87	2.165	2.130	0.390
4,530.00	87.97	131.11	2,726.43	-1,555.21	2,391.93	2,847.80	1.332	1.320	0.180
4,540.00	88.09	131.04	2,726.78	-1,561.78	2,399.47	2,857.74	0.417	0.360	-0.210
4,550.00	88.00	130.97	2,727.12	-1,568.33	2,407.01	2,867.67	0.342	-0.270	-0.210
4,560.00	86.93	130.50	2,727.56	-1,574.85	2,414.58	2,877.60	3.505	-3.210	-1.410

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Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
4,570.00	85.77	130.03	2,728.20	-1,581.30	2,422.19	2,887.54	3.754	-3.480	-1.410
4,580.00	85.83	130.08	2,728.93	-1,587.72	2,429.83	2,897.47	0.234	0.180	0.150
4,590.00	85.92	130.28	2,729.65	-1,594.16	2,437.45	2,907.39	0.657	0.270	0.600
4,600.00	85.76	130.48	2,730.38	-1,600.62	2,445.05	2,917.32	0.767	-0.480	0.600
4,610.00	85.73	130.66	2,731.12	-1,607.10	2,452.62	2,927.24	0.546	-0.090	0.540
4,620.00	85.92	130.80	2,731.85	-1,613.61	2,460.18	2,937.16	0.707	0.570	0.420
4,630.00	86.12	130.97	2,732.54	-1,620.14	2,467.72	2,947.08	0.787	0.600	0.510
4,640.00	85.71	131.05	2,733.25	-1,626.69	2,475.25	2,956.99	1.253	-1.230	0.240
4,650.00	85.63	131.17	2,734.01	-1,633.24	2,482.76	2,966.90	0.432	-0.240	0.360
4,660.00	85.80	131.28	2,734.75	-1,639.81	2,490.26	2,976.81	0.607	0.510	0.330
4,670.00	85.86	131.29	2,735.48	-1,646.39	2,497.75	2,986.71	0.182	0.180	0.030
4,680.00	85.42	131.21	2,736.24	-1,652.97	2,505.25	2,996.62	1.342	-1.320	-0.240
4,690.00	85.22	131.21	2,737.06	-1,659.54	2,512.75	3,006.52	0.600	-0.600	0.000
4,700.00	84.70	131.11	2,737.94	-1,666.09	2,520.25	3,016.42	1.588	-1.560	-0.300
4,710.00	83.90	130.82	2,738.93	-1,672.62	2,527.76	3,026.31	2.551	-2.400	-0.870
4,720.00	83.23	130.45	2,740.05	-1,679.09	2,535.30	3,036.20	2.293	-2.010	-1.110
4,730.00	82.81	130.25	2,741.27	-1,685.51	2,542.87	3,046.07	1.394	-1.260	-0.600
4,740.00	82.87	130.18	2,742.51	-1,691.92	2,550.44	3,055.95	0.275	0.180	-0.210
4,750.00	82.45	130.03	2,743.79	-1,698.31	2,558.03	3,065.82	1.337	-1.260	-0.450
4,760.00	82.23	129.83	2,745.12	-1,704.67	2,565.63	3,075.69	0.888	-0.660	-0.600
4,764.09	82.20	129.81	2,745.68	-1,707.26	2,568.74	3,079.73	0.264	-0.220	-0.147
4,777.63	80.87	130.20	2,747.67	-1,715.87	2,579.00	3,093.06	3.068	-2.947	0.864
4,817.95	77.41	129.45	2,755.27	-1,741.23	2,609.41	3,132.50	2.632	-2.574	-0.558
4,858.43	73.57	129.36	2,765.41	-1,766.10	2,639.68	3,171.55	2.847	-2.846	-0.067
4,894.35	69.40	129.21	2,776.81	-1,787.67	2,666.04	3,205.49	3.485	-3.483	-0.125
4,936.10	64.30	129.21	2,793.22	-1,811.93	2,695.77	3,243.75	3.665	-3.665	0.000
4,977.55	58.44	130.74	2,813.07	-1,835.28	2,723.65	3,279.96	4.351	-4.241	1.107
5,017.91	54.86	131.27	2,835.26	-1,857.39	2,749.09	3,313.46	2.681	-2.661	0.394
5,058.28	54.24	131.85	2,858.67	-1,879.21	2,773.70	3,346.11	0.579	-0.461	0.431
5,098.60	54.66	131.86	2,882.11	-1,901.10	2,798.13	3,378.66	0.313	0.312	0.007
5,138.97	54.49	132.53	2,905.51	-1,923.19	2,822.50	3,411.28	0.425	-0.126	0.498
5,179.34	54.59	131.33	2,928.93	-1,945.16	2,846.96	3,443.90	0.730	0.074	-0.892
5,219.44	54.66	130.59	2,952.15	-1,966.60	2,871.66	3,476.40	0.454	0.052	-0.554
5,259.72	54.14	131.17	2,975.60	-1,988.03	2,896.42	3,508.96	0.523	-0.387	0.432
5,300.07	54.61	130.13	2,999.10	-2,009.40	2,921.30	3,541.58	0.719	0.349	-0.773
5,331.00	54.61	130.13	3,017.01	-2,025.65	2,940.58	3,566.68	0.000	0.000	0.000

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F-10	Database:	Production EDM P246N

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
Theta Sør T#2.3 MM - actual wellpath misses target center by 396.17m at 5331.00m MD (3017.01 TVD, -2025.65 N, 2940.58 E) - Point	0.00	0.00	2,922.90	-2,340.34	3,162.08	6,476,224.00	438,211.00	58° 25' 15.8283 N	1° 56' 32.0767 E
Theta Sør T#2.2 MM 13l - actual wellpath misses target center by 55.16m at 5300.07m MD (2999.10 TVD, -2009.40 N, 2921.30 E) - Point	0.00	0.00	2,971.90	-2,057.24	2,925.00	6,476,507.00	437,974.00	58° 25' 24.8560 N	1° 56' 17.1984 E
Theta Sør Prelim 1 1606 - actual wellpath misses target center by 79.26m at 5255.75m MD (2973.27 TVD, -1985.91 N, 2893.99 E) - Point	0.00	0.00	2,979.90	-1,923.19	2,942.00	6,476,641.00	437,991.00	58° 25' 29.1964 N	1° 56' 18.1156 E
Theta Sør T#2.1 MM 13l - actual wellpath misses target center by 11.40m at 5265.42m MD (2978.93 TVD, -1991.07 N, 2899.90 E) - Point	0.00	0.00	2,979.90	-1,999.22	2,891.99	6,476,565.00	437,941.00	58° 25' 26.7140 N	1° 56' 15.1085 E
E TS - actual wellpath misses target center by 514.08m at 3812.49m MD (2675.21 TVD, -1147.00 N, 1807.48 E) - Point	0.00	0.00	3,064.90	-843.82	1,950.66	6,477,720.00	437,000.00	58° 26' 3.5664 N	1° 55' 15.9815 E
Theta Sør DRAFT Tgt - actual wellpath misses target center by 442.59m at 3657.75m MD (2664.39 TVD, -1069.55 N, 1674.00 E) - Point	0.00	0.00	3,070.90	-1,199.94	1,557.24	6,477,364.00	436,606.72	58° 25' 51.8537 N	1° 54' 52.0946 E
F-X VS - actual wellpath misses target center by 2172.38m at 2743.01m MD (2381.51 TVD, -620.91 N, 963.81 E) - Point	0.00	0.00	3,114.90	-1,748.13	-742.28	6,476,816.00	434,308.00	58° 25' 32.9179 N	1° 52' 30.9848 E
E VS - actual wellpath misses target center by 1951.72m at 2582.62m MD (2268.45 TVD, -554.03 N, 871.79 E) - Point	0.00	0.00	3,133.90	-1,278.97	-720.27	6,477,285.00	434,330.00	58° 25' 48.0905 N	1° 52' 31.8572 E
Volve Sør Injector 1201C - actual wellpath misses target center by 1951.72m at 2582.62m MD (2268.45 TVD, -554.03 N, 871.79 E) - Point	0.00	0.00	3,133.90	-1,278.97	-720.27	6,477,285.00	434,330.00	58° 25' 48.0905 N	1° 52' 31.8572 E
E W2 - actual wellpath misses target center by 2355.14m at 1930.00m MD (1803.72 TVD, -289.35 N, 498.16 E) - Point	0.00	0.00	3,264.90	838.77	-964.36	6,479,402.00	434,086.00	58° 26' 56.3915 N	1° 52' 14.6281 E

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
201.70	201.70	30"	30.000	36.000	
1,388.90	1,330.36	20"	20.000	26.000	
2,607.50	2,286.11	13 3/8"	13.375	17.500	
3,441.00	2,649.45	9 5/8"	9.625	12.250	

Checked By: _____	Approved By: _____	Date: _____
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